

**This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of
Bayesian ex-Gaussian Models in brms**

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Abstract

The ex-Gaussian distribution is a popular mathematical model for analyzing response time (RT) data because it separates the central portion of the distribution, captured by the Gaussian parameters μ and σ , from the positively skewed tail, captured by the exponential parameter τ . Researchers often use these parameters to make inferences about underlying cognitive processes. As Bayesian methods have become increasingly accessible, complex ex-Gaussian models can now be fit with relative ease, and in R, the `{brms}` package makes this especially straightforward. However, it is important to recognize that the default parameterization offered by `{brms}` does not map onto the classical parameterization familiar to many researchers in experimental psychology in the default model, μ indexes the mean of the full ex-Gaussian distribution rather than the location of the Gaussian component alone. This distinction matters because changes in the Gaussian location and changes in the exponential tail can offset one another at the level of the overall mean, so some published studies may have drawn incorrect inferences from effects estimated on the default μ parameter. Using simulated data and a published finding we highlight how using the default parameterization can lead to incorrect inferences. We then demonstrate how to fit the classical parameterization directly in `{brms}`, with the aim of helping researchers and reviewers align ex-Gaussian model parameters with their theoretical questions.

Keywords: ex-Gaussian, reaction time, Bayesian estimation, brms, parameterization

This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of Bayesian ex-Gaussian Models in brms

Over the past two decades, psychology has seen a substantial increase in the application of Bayesian data analysis methods as a supplement to widely used frequentist methods (Pfadt et al., 2025). This shift has been driven by the increasing availability of computational tools that have made fitting Bayesian models both practical and efficient. Programming languages like R (R Core Team, 2024), Python (Van Rossum & Drake, 2009), and Julia (Bezanson et al., 2017) provide libraries and packages that allow researchers to fit Bayesian models with little technical overhead. In addition, growing dissatisfaction with null-hypothesis significance testing (NHST) and its associated pitfalls (e.g., p-value misinterpretation and dichotomous thinking), together with the replication crisis/credibility revolution (Vazire, 2018), has pushed many researchers toward statistical alternatives.

One domain that has seen increased adoption of Bayesian methods is experimental psychology, where one of the most common dependent variables is response time (RT) data, which is a measure of how long it takes a person to initiate some type of response (e.g., motor, written, or speech). RT data offers a window into cognitive processes underlying behavior (Baayen & Milin, 2010). Traditionally, RT data have been analyzed using summary statistics such as the mean or median. However, RT distributions are typically positively skewed, bounded at zero, and characterized by substantial variability both within and between individuals (Whelan, 2008; Wilcox & Keselman, 2003). Moreover, experimental manipulations may affect not only the central tendency of the distribution but also its variability or upper tail. Consequently, analyses based solely on the mean or median can obscure theoretically meaningful distributional effects, motivating calls to move beyond mean-based analyses of RT data (Balota & Yap, 2011).

One way to move beyond the mean is to model the full shape of the RT distribution. Mathematical models provide a flexible way to do so, allowing researchers to draw inferences about the psychological processes—such as memory, attention, and language—that give rise to observed response times. One model that has become especially popular for this purpose is the ex-Gaussian model (Ratcliff & Murdock, 1976). The ex-Gaussian distribution is formed by the convolution of a Gaussian distribution and an exponential distribution and is characterized by three parameters: μ , σ , and τ . The parameter μ represents the mean of the Gaussian component, whereas σ represents its standard deviation. The τ parameter represents both the mean and standard deviation of the exponential component and primarily governs the length of the distribution's positively skewed upper tail. The overall mean of the ex-Gaussian

distribution is $\mu + \tau$, and the standard deviation (SD) is $\sqrt{\sigma^2 + \tau^2}$. Although these parameters have been used to infer cognitive processes such as automaticity (Balota et al., 2008; Dewitt et al., 2018) and cognitive control/inhibition (Kane & Engle, 2003), these interpretations are not inherent to the parameters and should be made cautiously (Matzke & Wagenmakers, 2009). Nevertheless, the ex-Gaussian model can provide a more complete description of RT data than the arithmetic mean alone, which can be disproportionately influenced by the positively skewed tail of the distribution.

The ex-Gaussian model can be fit within a Bayesian framework relatively easily using the {brms} package in R which provides a high-level interface to the probabilistic programming language Stan (Carpenter et al., 2017). The {brms} package supports a wide range of response distributions, including the ex-Gaussian family, and allows researchers to incorporate complex model structures at both the participant and item levels as well as across different parameters. This accessibility has contributed to the growing use of Bayesian ex-Gaussian models in reaction-time research (see Angele et al., 2023, 2024; Geller et al., 2025; Kinoshita & Liong, 2023; Serrano-Carot et al., 2026). However, an important complication arises from the default parameterization used by the R package {brms}. Although the model is straightforward to estimate, its default specification does not necessarily provide direct access to the parameter-specific comparisons researchers often wish to make.

The central problem is that the parameter labeled μ in the default {brms} ex-Gaussian model does not correspond to the mean of the Gaussian component in the conventional parameterization. Instead, the quantity modeled as μ in {brms} represents the mean of the full ex-Gaussian distribution. Under the conventional parameterization, this quantity is equal to $\mu_{\text{Gaussian}} + \tau$. Consequently, regression effects estimated for the default μ parameter reflect changes in a composite quantity that combines the Gaussian location and exponential tail components. Such effects therefore cannot be interpreted directly as changes in the Gaussian component or in typical response speed. To recover the conventional Gaussian mean, it is necessary to derive it from the model parameters as $\mu_{\text{Gaussian}} = \mu_{\text{brms}} - \tau$ or to specify an alternative parameterization that models the conventional ex-Gaussian parameters directly.

In this commentary, we demonstrate what the default parameterization of the ex-Gaussian model in {brms} actually provides and then show how to specify the model so that its parameters correspond to the conventional ex-Gaussian parameters used in reaction-time research. The goal is to improve the accuracy and interpretability of inferences drawn from these models. This issue is particularly important because the ex-Gaussian distribution is widely used to distinguish changes in the central portion of the RT distribution from changes in its positively skewed tail.

Simulation Example

To demonstrate the issue with fitting the default ex-Gaussian model in {brms}, I simulate data using a comparison adapted from one of my own studies (Geller et al., 2025) in R [version 4.6.0; R Core Team (2024)]. In Geller et al. (2025), we examined how perceptual disfluency—manipulated by varying the level of blur applied to letter strings influenced reaction times in a lexical decision task (i.e., is this letter string a word?), and whether these effects were related to how well someone remembers the words in a recognition memory task. In that study, low blur words affected early stages of encoding, reflected in the Gaussian location parameter μ , but not the later, tail component captured by τ . For the present illustration, however, I construct a sharper scenario in which μ and τ differ in opposite directions so that the two conditions share the same expected reaction time. Using the `rexgaussian2()` function from the {brms.exgaussian} package (installed from GitHub as shown in Listing 1), I simulate ex-Gaussian data for clearly presented and low-blur words in which μ is set to 500 ms and 550 ms, τ to 100 ms and 50 ms, and σ to 60 ms in both conditions (see Table 1).

Listing 1

Installing and loading the {brms.exgaussian} package from GitHub.

```
# install from Github
#remotes::install_github("mattansb/brms.exgaussian")

library(brms.exgaussian)
```

Table 1

Data-generating ex-Gaussian parameters for the Clear and Low-Blur conditions, with opposing changes in μ and τ that hold the expected response time ($\mu + \tau$) constant.

Blur	μ	σ	τ	Expected RT
Clear	500	60	100	600
Low-Blur	550	60	50	600

Simulate the Data

We first draw 100 reaction times per condition (200 in total) from the ex-Gaussian distributions defined by these parameters (Listing 2).

Listing 2

Simulating ex-Gaussian reaction times for the Clear and Low-Blur conditions.

```
# Fix the random seed so the simulated data are identical on every render.
set.seed(1234)

# Number of simulated trials per condition (200 observations in total).
n_per_group <- 100

# One row per trial, labelled by Blur condition (Clear is the reference level).
d <- data.frame(
  Blur = factor(
    rep(c("Clear", "Low-Blur"), each = n_per_group),
    levels = c("Clear", "Low-Blur")
  )
)

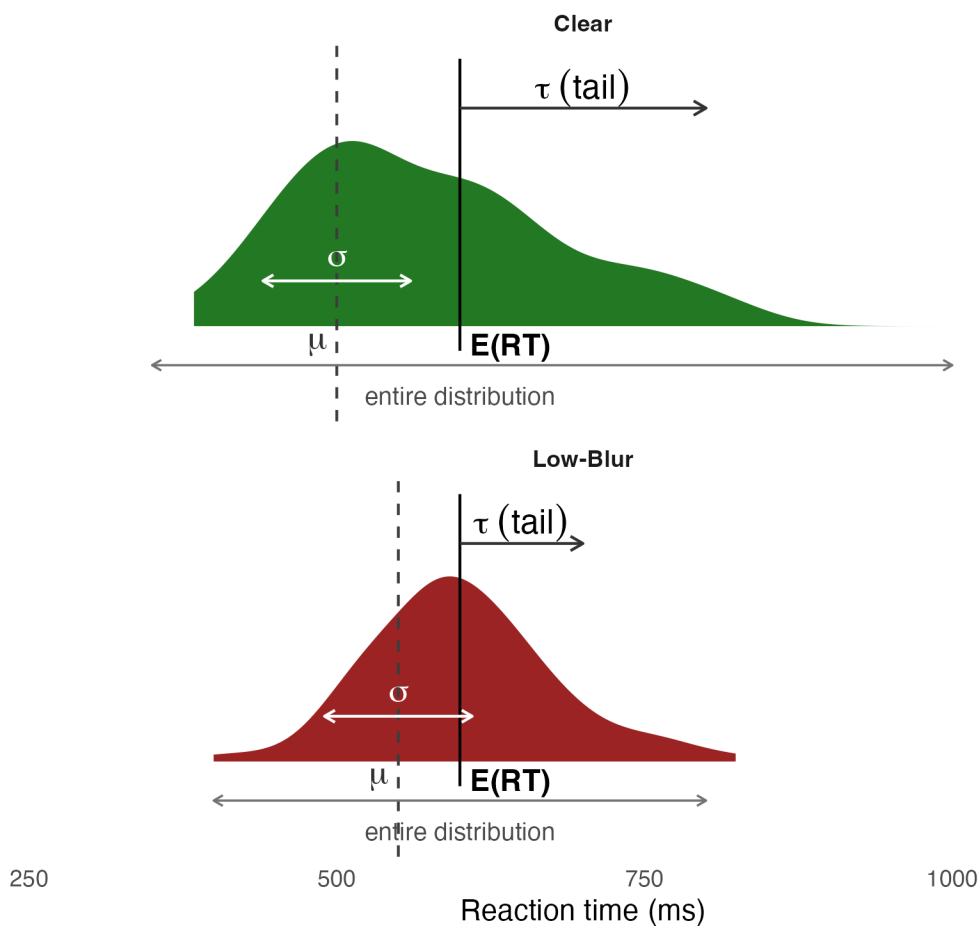
# Row index (1 = Clear, 2 = Low-Blur) used to look up each trial's parameters.
parameter_index <- as.integer(d$Blur)

# Draw reaction times from the ex-Gaussian using each trial's condition values.
d$rt <- brms.exgaussian::rexgaussian2(
  n = nrow(d),
  mu = sim_params$mu[parameter_index],
  sigma = sim_params$sigma[parameter_index],
  tau = sim_params$tau[parameter_index]
)
```

Figure 1 shows the simulated RT distribution for each condition and how the three ex-Gaussian parameters shape it: μ locates the Gaussian component, σ sets its spread, and τ stretches the right tail. Because the increase in μ from Clear to Low-Blur is offset by an equal decrease in τ , the overall mean $E(RT) = \mu + \tau$ is identical (600 ms) in both conditions.

Figure 1

Simulated reaction-time distributions for each condition, annotated with the ex-Gaussian parameters. The dashed line marks the Gaussian location μ ; the double arrow spans $\pm\sigma$ around it (the Gaussian spread); the rightward arrow over the tail marks the exponential component τ , which stretches the distribution to the right; and the solid line marks the overall mean $E(RT) = \mu + \tau$, which is 600 ms in both conditions. The bottom span highlights that $E(RT)$ is the mean of the entire distribution — the quantity the default brms mu estimates.

**Default {brms} Parameterization**

Using this data we fit the default brms ex-Gaussian model. In this parameterization, the parameter labeled mu represents the expected value of the full ex-Gaussian distribution rather than the location of its Gaussian component.

The exponential component is labeled beta. On the response scale, beta corresponds to the classical ex-Gaussian parameter τ :

$$\beta = \tau_{\text{classical}}.$$

174 Thus, β and τ are not substantively different quantities. Both represent the mean of the exponential
175 component. The difference is primarily one of notation. The important difference between the default {}
176 model and the classical parameterization concerns μ :

$$\mu = \mu_{\text{classical}} + \tau.$$

177 In other words, the default {} parameterization uses μ for the expected value of the complete ex-
178 Gaussian distribution, whereas the classical parameterization uses μ for the location of the Gaussian
179 component.

180 Both the expected response and the exponential component are allowed to vary across conditions.

181 ***Prior Specification***

182 The priors were selected to reflect the known scale of the simulated data while remaining
183 sufficiently broad to allow the model to recover deviations from the generating values. [Table 2](#) shows
184 these priors alongside those used for the classical parameterization introduced later, so the two
185 specifications can be compared directly.

Table 2*Prior specification for the default and classical ex-Gaussian models.*

Parameter	Default Prior	Classical Prior	Notes
Intercept (μ)	Normal(600, 100)	Normal(500, 100)	Same form; centered on each parameterization's reference-condition value (E(RT) = 600 ms vs. Gaussian location = 500 ms)
Blur (μ)	Normal(0, 50)	Normal(0, 50)	Identical across parameterizations; no assumed direction for the effect of Blur
Intercept (beta / tau, log scale)	Normal(log(100), 0.5)	Normal(log(100), 0.5)	Identical across parameterizations; centered at log(100 ms), the simulated exponential mean of the reference condition
Blur (beta / tau, log scale)	Normal(0, 0.5)	Normal(0, 0.5)	Identical across parameterizations; no assumed direction for the effect of Blur
Intercept (sigma, log scale)	brms default	Normal(log(60), 0.5)	Only the classical model estimates sigma explicitly; centered at log(60 ms), the simulated Gaussian standard deviation

The model can then be fit as follows (Listing 3):

Listing 3

Fitting the default ex-Gaussian model in brms.

```
fit_brms <- brm(
  bf(
    rt ~ Blur, # mu submodel: here mu is E(RT), not the Gaussian location
    sigma ~ 1, # constant Gaussian SD
    beta ~ Blur # exponential mean (the classical tau), allowed to vary
  ),
  data = d,
  prior = priors_default,
  family = exgaussian(), # 's built-in ex-Gaussian family
  chains = 2,
  iter = 2000,
  warmup = 500,
  backend = "cmdstanr",
  cores = 4,
  seed = 1234 # reproducible draws
)
```

Estimating Ex-Gaussian Parameters with {easystats}

The {modelbased} package, which is part of the {easystats} ecosystem, can easily be used to obtain condition-specific posterior estimates from a fitted model. The `estimate_means()` function returns estimated parameters for each condition, back-transformed, so estimates are presented on time scale, whereas `estimate_contrasts()` computes differences between conditions.

As shown in [Table 1](#), the present data were generated so that the two conditions have the same expected reaction time but differ in the Gaussian and exponential components of the ex-Gaussian distribution.

Under the classical ex-Gaussian parameterization, the expected response is

$$E(RT) = \mu + \tau.$$

Thus, the 50-ms increase in μ for the Low-Blur condition is offset by a 50-ms decrease in τ . The expected RT is therefore 600 ms in both conditions.

Estimating the Expected Response (μ)

Condition-specific estimates of the expected response can be obtained with `estimate_means()` and setting `predict` argument to “mu” (Listing 4):

Listing 4

Estimating the condition-specific expected response (μ) from the default model.

```
# Default model: per-condition mu, which equals E(RT) in this parameterization.
overall_means <- estimate_means(
  fit_brms,
  by = "Blur",
  predict = "mu",
  centrality = "mean"
)
```

Table 3

Estimated expected response, $E(RT)$, by condition from the default model.

Blur	Mu	95% CrI	pd
Clear	580.8	[559.1, 603.8]	1
Low-Blur	603.2	[588, 618.9]	1

Because the primary μ parameter in the default ex-Gaussian model represents $E(RT)$, these estimates should be close to 600 ms in both conditions.

The corresponding contrast can be obtained using `estimate_contrasts()` (Listing 5):

Listing 5

Estimating the Blur contrast in the expected response (default model).

```
# Default model: Blur contrast in E(RT) ( $\approx 0$  by design).
overall_difference <- estimate_contrasts(
  fit_brms,
  contrast = "Blur",
  centrality = "mean"
)
```

Table 4

Blur contrast in the expected response from the default brms model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	22.4	[-4.8, 48.8]	0.949

The posterior estimate of this contrast should be close to zero because the two conditions were generated with the same expected RT:

$$(550 + 50) - (500 + 100) = 600 - 600 = 0.$$

Thus, the default model correctly recovers the absence of a difference in the full distributional mean. However, its primary μ coefficient does not directly reveal the 50-ms difference in the Gaussian-location parameter.

Estimating the Exponential Component.

In the default {brms} parameterization, β represents the mean of the exponential component. It therefore corresponds directly to τ in the classical parameterization:

$$\beta = \tau.$$

Condition-specific estimates can be obtained by requesting predictions for β (Listing 6):

Listing 6

Estimating the condition-specific exponential component (beta) from the default model.

```
# Default model: per-condition beta (the exponential mean = classical tau).
tau_estimates_default <- estimate_means(
  fit_brms,
  by = "Blur",
  predict = "beta",
  centrality = "mean"
)
```

Table 5

Estimated exponential component (beta) by condition from the default brms model.

Blur	Beta	95% CrI	pd
Clear	107.3	[79.1, 139.6]	1
Low-Blur	55.7	[34.9, 79.5]	1

Although the resulting object is called `tau_estimates_default` for interpretive clarity, the parameter is named `beta` inside the default brm model.

These estimates should be approximately 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using (Listing 7):

Listing 7

Estimating the Blur contrast in the exponential component (beta) from the default model.

```
# Default model: Blur contrast in beta (the exponential component).
tau_difference_default <- estimate_contrasts(
  fit_brms,
  contrast = "Blur",
  predict = "beta",
  centrality = "mean"
)
```

Table 6

Blur contrast in the exponential component (beta) from the default brms model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	-51.6	[-86.1, -19.7]	0.999

As we can see in [Table 6](#), the difference between the two conditions is approximately 50 ms.

Classical Ex-Gaussian Parameterization

As we have shown above, the default parameterization gives you the expected RT of the entire distribution. We can instead fit the model using the `{brms.exgaussian}` package, which implements the classical ex-Gaussian parameterization. In this model, μ represents the location of the Gaussian component and τ represents the mean of the exponential component.

We first define the custom family and Stan functions from the `{brms.exgaussian}` package (Listing 8):

Listing 8

Defining the custom classical ex-Gaussian family and Stan code.

```
# Custom family that reparameterizes brms so mu is the Gaussian location.
exg_stanvars <- exgaussian2_stancode() # Stan code injected into the model
exg_family <- exgaussian2() # the custom family definition
```

Prior Specification

The priors for the classical model follow the same reasoning as those for the default model, shown together in [Table 2](#):

The model is then fit using (Listing 9):

Listing 9

Fitting the classical ex-Gaussian model using {brms.exgaussian}.

```
fit_brms_new <- brm(
  bf(
    rt ~ Blur, # mu submodel: here mu IS the Gaussian location
    tau ~ Blur, # exponential mean (the tail), allowed to vary
    sigma ~ 1 # constant Gaussian SD
  ),
  data = d,
  prior = priors_classical,
  family = exg_family, # custom classical ex-Gaussian family
  stanvars = exg_stanvars, # supplies its Stan implementation
  chains = 2,
  iter = 2000,
  warmup = 500,
  backend = "cmdstanr",
  cores = 4,
  seed = 1234
)
```

Estimating the Gaussian-Location Parameter

For the classical model, estimates from the primary regression formula correspond to the Gaussian-location parameter μ (Listing 10):

Listing 10

Estimating the Gaussian-location parameter (μ) from the classical model.

```
# Classical model: per-condition mu, now the Gaussian-location parameter.
mu_estimates <- estimate_means(
  fit_brms_new,
  by = "Blur",
  predict = "mu",
  centrality = "mean"
)
```

Table 7

Estimated Gaussian-location parameter (μ) by condition from the classical model.

Blur	Mu	95% CrI	pd
Clear	476.7	[452.1, 503.3]	1
Low-Blur	544.7	[524.5, 564.7]	1

The estimates are approximately 500 ms for Clear and 550 ms for Low-Blur.

The contrast between the two can be obtained using (Listing 11):

Listing 11

Estimating the Blur contrast in the Gaussian-location parameter (classical model).

```
# Classical model: Blur contrast in the Gaussian-location parameter mu.
mu_difference <- estimate_contrasts(
  fit_brms_new,
  contrast = "Blur",
  predict = "mu",
  centrality = "mean"
)
```

Table 8

Blur contrast in the Gaussian-location parameter (μ) from the classical model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	68	[39.2, 95.8]	1

The posterior mean will not necessarily equal exactly 50 ms because of sampling variability. However, the 95% credible interval should contain the population value of 50 ms when the model successfully recovers the simulated effect.

Estimating the Exponential Component

Condition-specific estimates of τ can be obtained by requesting predictions for the tau distributional parameter (Listing 12):

Listing 12

Estimating the exponential component (tau) from the classical model.

```
# Classical model: per-condition tau (the exponential component).
tau_estimates <- estimate_means(
  fit_brms_new,
  by = "Blur",
  predict = "tau",
  centrality = "mean"
)
```

Table 9

Estimated exponential component (τ) by condition from the classical model.

Blur	Mean	95% CrI	pd
Clear	104.0	[76.2, 136.2]	1
Low-Blur	58.2	[38.5, 81.6]	1

The estimates should be close to 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using (Listing 13):

Listing 13

Estimating the Blur contrast in the exponential component (τ) from the classical model.

```
# Classical model: Blur contrast in tau (the exponential component).
tau_difference <- estimate_contrasts(
  fit_brms_new,
  contrast = "Blur",
  predict = "tau",
  centrality = "mean"
)
```

Table 10

Blur contrast in the exponential component (τ) from the classical model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	-45.8	[-77.3, -14]	0.997

The results seen in [Table 10](#) are very similar to the default parameterization.

This example illustrates the central difference between the two parameterizations. The default `{brms}` ex-Gaussian model directly estimates the expected response and labels that parameter μ . The exponential component is labeled β , although it corresponds to the classical τ parameter. In contrast, the `{brms.exgaussian}` model uses μ for the Gaussian-location parameter and τ for the exponential mean, consistent with the classical parameterization commonly used in reaction-time research.

Real Example

Conclusion

With this commentary, I hope to encourage researchers to use the appropriate ex-Gaussian parameterization when modeling reaction-time data with the `{brms}` package. The ex-Gaussian distribution provides a powerful framework for moving beyond analyses of the mean by separating differences in the central portion of the reaction-time distribution from differences in its right tail.

However, this flexibility also requires careful attention to model parameterization. In particular, researchers should not assume that parameters with familiar labels necessarily have the same interpretation across software implementations. Researchers should therefore examine the statistical definition and underlying implementation of a model before fitting it and interpreting its parameters. This is especially important when using flexible modeling software that may adopt parameterizations that differ from those commonly used in a particular research literature. I hope this commentary serves both as

a practical guide to fitting the classical ex-Gaussian model in {brms} and as a broader reminder that convenient software defaults should not replace careful consideration of what a model actually estimates.

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